

1. **Experiment:** Preparation of Fluorescent Complex and Flow Cytometric Detection of Complex Internalization
2. **Time:** 2026.01.18-2026.01.24
3. **Member:** Yang Jin, Binxuan Zhang, Kaiqing Zhang, Zheyuan Deng, Yuxuan Wang
4. **Materials:** Recombinant human sIL-6R, recombinant human IL-6, EZ-Link™ Sulfo-NHS-Biotinylation Kit (Thermo Scientific, Cat. No. 21425), PBS buffer, Streptavidin-APC (BD Biosciences, Cat. No. 554067), Raji cell library, pre-chilled PBS, unstained cell control, 4 °C incubation control.
5. **Method:**
 - (1) Preparation of fluorescently labeled IL-6/sIL-6R α complex
 - ① Recombinant human sIL-6R and IL-6 were mixed at a 1:1 molar ratio and incubated at room temperature for 30 min to form the IL-6/sIL-6R α complex.
 - ② Biotinylation was performed using the EZ-Link™ Sulfo-NHS-Biotinylation Kit (Thermo Scientific, Cat. No. 21425). The complex was buffer-exchanged into PBS and then incubated with a 20x molar excess of Sulfo-NHS-Biotin reagent for 30 min at room temperature in the dark, allowing covalent conjugation of biotin to primary amine groups via stable amide bonds. The sample was centrifuged at 14,000 rpm for 15 min at 4 °C and the supernatant was collected.
 - ③ After the reaction, excess free biotin was removed using Zeba Spin desalting columns to obtain purified biotinylated complex.
 - ④ The biotinylated complex was then incubated with Streptavidin-APC (BD Biosciences, Cat. No. 554067) at a molar ratio of 1:4 for 30 min at 4 °C in the dark to obtain the fluorescently labeled IL-6/sIL-6R α complex.
 - (2) intracellular APC fluorescence assay
 - ① The Raji cell library was incubated with the fluorescently labeled complex at 37 °C for 4 h (final complex concentration: 50 nM).
 - ② Cells were washed twice with pre-chilled PBS to terminate internalization and remove unbound fluorescent complexes.
 - ③ Cells were sorted by flow cytometry based on intracellular APC fluorescence intensity. Gating boundaries were established using unstained cells and a 4 °C incubation control (internalization inhibited) as references, and cell populations with APC fluorescence intensity in the top 5% were collected.
6. **Result:**
 - (1) Following co-incubation with the fluorescently labeled complex, flow cytometry was performed to detect intracellular APC fluorescence. Staining was gated with reference to unstained cells and a 4 °C incubation control (internalization inhibited). Cells with APC fluorescence intensity in the top 5% were collected as positive population.
 - (2) After three rounds of FACS sorting with decreasing complex concentrations (50 nM $\rightarrow$ 10 nM $\rightarrow$ 2 nM), the percentage of APC-positive cells progressively increased, indicating successful enrichment of Raji cell clones capable of binding and internalizing the IL-6/sIL-6R α complex.
 - (3) The sorted APC-positive cells were expanded and processed for genomic DNA extraction and third-generation sequencing.

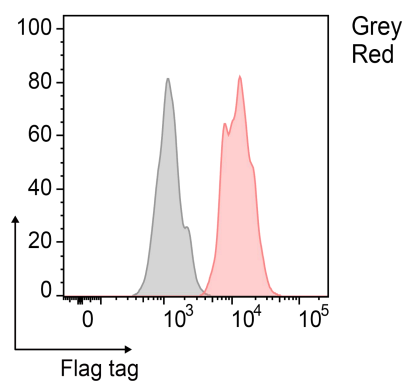


Figure 1. Representative flow-cytometry histograms of intracellular APC fluorescence in Raji cells incubated with the fluorescently labeled IL-6/sIL-6R α complex. Grey, unstained cells and the 4 °C internalization-inhibited control; red, Raji cells incubated with the fluorescently labeled complex at 37 °C.